

Design and Implementation of a Smart Solar Energy Management and Water Heating System Based on Dual- Axis Tracking and Internet of Things (IoT)

Graduation Project II Report Submitted to the Department of Control
Engineering and Robotics in Partial Fulfillment of the Requirements for the
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Dedication

I dedicate this work to my family, whose endless support, encouragement, patience, and belief in me have been the true foundation behind every step of this achievement. Their motivation and sacrifices have always inspired me to continue striving toward my goals.

I would like to express my sincere gratitude and appreciation to my supervisor, Eng. Marwa Al-Safadi, for her valuable guidance, continuous support, constructive feedback, and constant encouragement throughout all stages of this project.

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Finally, I dedicate this work to everyone who supported me, encouraged me, and stood beside me throughout this journey.

Supervisor's Declaration

I hereby certify that the preparation of this project entitled:

“Design and Implementation of a Smart Solar Energy Management and Water Heating System Based on Dual-Axis Tracking and Internet of Things (IoT)”

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Abstract

This project presents the design and implementation of an integrated smart system for solar energy management and water heating, based on dual-axis tracking and Internet of Things (IoT) technologies. The system is developed to enhance energy efficiency, ensure operational reliability, and provide real-time monitoring and control.

The proposed system consists of several interconnected subsystems. A dual-axis solar tracking mechanism dynamically orients the photovoltaic panel toward the maximum solar irradiance, thereby maximizing energy capture and improving overall conversion efficiency throughout the day. The generated energy is stored within a stable and integrated electrical storage system designed to ensure continuous operation and minimize the impact of environmental fluctuations. A closed-loop thermal control system continuously monitors the water temperature inside the tank and regulates the heating process according to a predefined reference value, ensuring optimal thermal performance with minimal energy loss.

In addition, the system incorporates a wireless control and monitoring interface via a smart application, enabling users to remotely adjust temperature settings and observe system status in real time. Safety is enhanced through a water level monitoring system that prevents dry operation, as well as a leakage detection mechanism that triggers immediate protective actions, including water supply isolation and user alerts. A local display interface is also provided to present key operational parameters such as current temperature, target values, and energy status.

The project addresses several key challenges, including inefficient utilization of solar energy, reliance on conventional energy sources for water heating, limited capability for real-time monitoring of temperature and water levels, and the absence of early leak detection systems. Accordingly, the main objectives are to improve energy and water efficiency, reduce dependence on traditional electrical heating, enable intelligent remote monitoring and control, and enhance system safety through automated protection mechanisms.

Overall, the system demonstrates a practical and scalable approach toward sustainable energy utilization and smart resource management.

Keywords

Smart Solar System, Dual-Axis Tracking, Water Heating, Internet of Things (IoT), Thermal Control, Energy Efficiency, Remote Monitoring, Safety System.

الملخص

يقدم هذا المشروع تصميم وتنفيذ منظومة ذكية متكاملة لإدارة الطاقة الشمسية وتسخين المياه، بالاعتماد على تقنية التتبع الشمسي ثنائي المحور وتقنيات إنترنت الأشياء (IoT)، بهدف تحسين كفاءة استخدام الطاقة، وضمان استمرارية التشغيل، وتوفير مراقبة وتحكم لحظي في النظام.

تتألف المنظومة من عدة أنظمة فرعية مترابطة، حيث تعتمد على آلية تتبع شمسي ثنائي المحور تقوم بتوجيه اللوح الشمسي تلقائياً نحو أعلى شدة إشعاع شمسي، مما يساهم في زيادة القدرة المستخرجة وتحسين كفاءة التحويل الطاقوي خلال مختلف أوقات النهار. يتم تخزين الطاقة المنتجة ضمن نظام تخزين كهربائي متكامل يضمن استقرار التغذية واستمرارية عمل النظام حتى في حالات انخفاض الإشعاع الشمسي. كما تتضمن المنظومة نظام تحكم حراري مغلق الحلقة يقوم بمراقبة درجة حرارة المياه داخل الخزان بشكل مستمر، والتحكم بعملية التسخين وفق قيمة مرجعية محددة مسبقاً، بما يحقق الكفاءة الحرارية المطلوبة مع تقليل الفاقد الطاقوي.

تدعم المنظومة واجهة تحكم ومراقبة لاسلكية من خلال تطبيق ذكي يتيح للمستخدم ضبط درجة الحرارة ومتابعة الحالة التشغيلية بشكل لحظي وعن بُعد. كما تم تعزيز عناصر الأمان عبر نظام لمراقبة مستوى المياه يمنع التشغيل الجاف، بالإضافة إلى نظام كشف تسرب مائي يعمل على إصدار تنبيهات فورية واتخاذ إجراءات وقائية مثل عزل مصدر المياه. كذلك، توفر المنظومة واجهة عرض محلية تعرض القيم التشغيلية الأساسية مثل درجة الحرارة الحالية، القيمة المطلوبة، وحالة الطاقة.

يعالج المشروع مجموعة من التحديات الأساسية، من أبرزها انخفاض كفاءة استغلال الطاقة الشمسية، والاعتماد على مصادر الطاقة التقليدية في تسخين المياه، وصعوبة المراقبة اللحظية لمستوى المياه ودرجة الحرارة، بالإضافة إلى غياب أنظمة فعالة للكشف المبكر عن التسربات. وعليه، يهدف المشروع إلى تقليل الهدر المائي والطاقوي، وتعزيز الاستدامة البيئية، وتمكين التحكم والمراقبة عن بُعد، إضافة إلى تحسين كفاءة النظام وتوفير آليات حماية تلقائية تضمن سلامة التشغيل.

بشكل عام، يمثل هذا المشروع نموذجاً عملياً لمنظومات الطاقة الذكية التي تدمج بين الطاقة المتجددة وتقنيات التحكم الحديثة، بما يساهم في تحقيق إدارة أكثر كفاءة واستدامة للموارد.

الكلمات المفتاحية

نظام شمسي ذكي، تتبع ثنائي المحور، تسخين المياه، إنترنت الأشياء، التحكم الحراري، كفاءة الطاقة، المراقبة عن بعد، نظام الأمان.

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List of Abbreviations

Abbreviation	Full Term
ESP	Espressif Microcontroller Platform
IoT	Internet of Things
ADC	Analog-to-Digital Converter reading
LDR	Light Dependent Resistor
I/O	Input / Output
DC	Direct Current
GPIO	General Purpose Input/Output
I2C	Inter-Integrated Circuit
SDA	Serial Data Line
SCL	Serial Clock Line
KPIs	Key Performance Indicators
PID	Proportional-Integral-Derivative

List of Symbols

Symbol	Description
Q	Heat energy
m	Mass of water
C	Specific heat capacity
ΔT	Temperature difference
T	Temperature
T	Time
P	Power
η	System efficiency
V	Voltage
I	Current
Θ	Angular position (servo angle)
E	Battery energy
Ah	Battery capacity
Wh	Watt-hour (energy unit)
W	Watt (power unit)
A	Ampere (current unit)
$^{\circ}\text{C}$	Temperature in Celsius

Chapter 1

General Introduction

1.1 Project Overview and Background

In recent years, there has been a growing global interest in utilizing renewable energy sources, particularly solar energy, due to its sustainability and environmental benefits. However, one of the major limitations of conventional solar systems is their relatively low efficiency, as solar panels are typically fixed at a constant angle, resulting in significant energy loss throughout the day.

To address this issue, solar tracking systems have been developed to continuously orient solar panels toward the sun using light sensors and motorized mechanisms, thereby maximizing solar radiation capture and improving overall system performance. Previous studies have demonstrated that solar tracking systems significantly increase energy output compared to fixed solar panels [1].

Based on this concept, the development of this project began with a single-axis solar tracking system implemented during the semester project, which was later enhanced into a dual-axis tracking system in Graduation Project 1 to achieve higher tracking accuracy. In the current stage, the system has been further extended to include a smart solar water heating application, integrated with Internet of Things (IoT) technology for real-time monitoring and remote control.

The proposed system is built around an ESP32 microcontroller, which integrates various sensors and actuators into a unified platform. Light Dependent Resistors (LDRs) are used to detect the direction of solar radiation and control the movement of the solar panel using servo motors. In addition, a DS18B20 digital temperature sensor is employed for accurate water temperature measurement, as it has been proven to provide reliable and stable readings in low-cost thermal monitoring applications [2]. Furthermore, the system incorporates water level and leakage detection sensors to ensure safe operation, as studies have shown that such systems enhance operational safety and reduce potential risks [3][4].

Accordingly, this project presents an integrated smart system that combines solar tracking, thermal control, and safety mechanisms within a single platform, aiming to improve solar energy utilization efficiency and provide a reliable solution for solar water heating applications.

1.2 Problem Statement and Importance

Despite the widespread adoption of solar water heating systems, many existing solutions still suffer from significant limitations in terms of efficiency and operational safety, which restrict their practical effectiveness. One of the primary challenges is the reliance on fixed solar panels that do not adapt to the sun's movement throughout the day, resulting in reduced solar energy capture and overall system efficiency. Previous studies have demonstrated that solar tracking systems can significantly enhance energy output compared to fixed systems [1].

In addition, conventional heating systems often lack precise thermal control mechanisms, where the heating element continues to operate without considering the required temperature limit, leading to unnecessary energy consumption. Furthermore, the absence of accurate temperature measurement negatively affects system performance, whereas studies have shown that digital sensors such as the DS18B20 provide reliable and stable temperature readings in thermal monitoring applications [2].

From a safety perspective, these systems present several operational risks. Heating may occur even when the water level is low or when the tank is empty, potentially causing system damage or hazardous conditions. Moreover, the lack of effective leakage detection mechanisms makes it difficult to identify faults at an early stage, although conductivity-based detection systems have been proven to provide a simple and efficient solution for detecting water leakage [4]. Similarly, water level

monitoring systems have been shown to improve system management and prevent overflow or shortage conditions [3].

Another critical limitation is the absence of intelligent monitoring capabilities, which prevents users from tracking system performance in real time, particularly when the system is installed in remote or elevated locations. This lack of monitoring reduces system reliability and increases the likelihood of undetected failures.

Accordingly, there is a clear need for an integrated solution that enhances solar energy utilization, ensures precise thermal control, improves operational safety, and enables real-time monitoring. Therefore, this project proposes a smart system that incorporates a dual-axis solar tracking mechanism to maximize solar energy capture, combined with a precise thermal control system, supported by reliable sensors and safety mechanisms including water level monitoring and leakage detection. In addition, an Internet of Things (IoT)-based platform is implemented to allow remote monitoring and control of the system in an efficient and user-friendly manner.

1.3 Objectives

The project objectives are defined based on measurable performance indicators to ensure a systematic evaluation of system efficiency, safety, and reliability, following the SMART criteria (Specific, Measurable, Achievable, Relevant, and Time-bound).

These objectives are based on well-established principles in previous studies, including solar tracking efficiency improvement [1], accurate temperature measurement [2], automated water level control [3], and leakage detection systems [4].

1.3.1 Main Objective

The main objective of this project is to design and implement an integrated smart system for solar energy management and water heating, based on dual-axis solar tracking and Internet of Things (IoT) technologies, in order to enhance solar energy utilization, reduce energy losses, and ensure safe and stable operation according to user-defined parameters.

1.3.2 Specific Objectives

1. Solar Tracking Performance

To achieve dual-axis solar tracking with an angular accuracy of approximately $\pm 5^\circ$ and a response time of 1–2 seconds, aiming to improve solar energy capture efficiency by an estimated 20–40% compared to fixed systems [1].

2. Thermal Heating Control

To heat water within a temperature range of 40–60°C, while maintaining temperature stability within $\pm 2^\circ\text{C}$ around the setpoint, using a DS18B20 digital temperature sensor known for its reliability and stability [2].

3. Real-Time Monitoring and IoT Control

To provide real-time monitoring with an update rate of ≤ 1 second, enabling remote control via a mobile application, along with local data display using an LCD interface.

4. Safety and Protection System

To ensure water leakage detection within ≤ 1 second using conductivity-based methods [4], and to prevent heater operation in case of low or absent water level based on monitoring techniques [3], in addition to automatic shutdown at excessive temperatures.

5. Energy Efficiency Optimization

To reduce electrical energy consumption by an estimated 20–30% through efficient solar tracking and eliminating unnecessary heating operation.

6. System Reliability and Stability

To ensure continuous system operation (24/7 monitoring) while minimizing oscillations in tracking movement and heater operation.

1.3.3 Mathematical Basis for Performance Evaluation

The thermal performance of the system is evaluated based on the amount of heat energy required to raise the water temperature, which is expressed as :

$$(1) Q = m * c * \Delta T$$

Where

- Q is the required thermal energy
- m is the mass of water
- c is the specific heat capacity
- ΔT is the temperature difference between the initial and target temperatures

The expected heating time is calculated based on the heater power using the following relation:

$$(2) \ t = Q \ / \ P$$

Where

- t represents the heating time
- P is the heater power

Based on these equations, the heating duration typically ranges between 2 to 6 hours depending on the water volume, temperature difference, heater power, and solar irradiance conditions, while also considering system thermal losses.

1.4 Definitions and Basic Concepts

This project is based on a set of fundamental concepts related to solar energy systems, control engineering, and Internet of Things (IoT) technologies, which form the theoretical foundation for the design and implementation of the proposed system. Understanding these concepts is essential for analyzing system behavior and evaluating its performance.

Solar energy refers to the energy derived from solar radiation, which can be converted into thermal or electrical energy and is widely used in renewable energy applications, particularly for water heating systems.

Solar tracking systems are used to automatically orient solar panels toward the sun in order to maximize solar radiation capture. Studies have shown that dual-axis tracking systems significantly improve energy efficiency compared to fixed systems [1].

In this project, the tracking mechanism is based on **Light Dependent Resistors (LDRs)**, which are sensors whose resistance varies according to light intensity, enabling the detection of solar direction.

The movement of the solar panel is controlled using **servo motors**, which provide precise angular positioning.

The system is managed by an **ESP32 microcontroller**, which offers both processing capabilities and wireless communication, enabling the integration of **Internet of Things (IoT)** technologies for real-time monitoring and remote control.

For thermal regulation, the system utilizes a **DS18B20 digital temperature sensor**, which provides accurate and stable temperature measurements and has been proven suitable for low-cost thermal monitoring applications [2].

The heating process is controlled through a **thermal control system**, which maintains the water temperature within a predefined range based on the user-defined **setpoint**.

To ensure safe operation, the system incorporates a **water level sensor** to monitor the tank condition and prevent unsafe operation, as water level monitoring systems have been shown to improve system management and safety [3].

In addition, a **leakage detection system** is implemented based on electrical conductivity principles, where the presence of water bridges conductive electrodes, enabling early detection of leaks [4].

Finally, system performance is evaluated based on **system efficiency**, which represents the ability of the system to convert available energy into useful output while minimizing energy losses.

1.5 Scope and Technical Requirements

1.5.1 Project Scope

This project aims to design and implement a smart embedded system for solar energy management and water heating, specifically targeted for residential applications. The system integrates dual-axis solar tracking, thermal control, and Internet of Things (IoT) technologies into a unified platform.

The system includes a dual-axis solar tracking mechanism to maximize solar radiation capture, along with a thermal control system based on a user-defined setpoint, where the heating process is automatically controlled (ON/OFF) according to the desired temperature. In addition, the system provides real-time monitoring through a mobile application connected via Wi-Fi, as well as local data visualization using an LCD display.

The system incorporates multiple sensors to measure light intensity, water temperature, water level, electrical voltage, and leakage conditions. These components are managed by an ESP32 microcontroller, which serves as the central processing unit responsible for decision-making and controlling actuators such as servo motors, the heating element, and the water pump.

Furthermore, the system is designed with a strong focus on safety by preventing operation under unsafe conditions, such as low or empty water levels, and by enabling early leakage detection, in addition to implementing thermal shutdown at the desired temperature.

However, this project does not include artificial intelligence or machine learning techniques, does not support grid connection, and is not intended for large-scale or industrial applications. It also excludes advanced features such as weather prediction or intelligent load optimization.

Accordingly, the system can be defined as a smart embedded energy management system for residential use, focusing on efficiency, safety, and real-time monitoring without introducing unnecessary complexity.

1.5.2 Technical Requirements

The technical requirements of the system are defined based on the intended performance objectives and include mechanical, thermal, electrical, and control aspects. The following table summarizes the key specifications of the system:

Table 1.1: Technical Specifications of the System

Section	Technical Specification	Value / Requirement	Verification Method
Solar Tracking System	Tracking Accuracy	Approximately $\pm 5^\circ$	Practical testing using LDR sensors
Solar Tracking System	Energy Gain vs Fixed Panel	20–40% improvement	Performance comparison with fixed panel (Literature-based)
Solar Tracking System	Tracking Mechanism	Dual-axis (2 Servo Motors + 4 LDR)	Hardware inspection
Solar Tracking System	Sensor Type	4× LDR (Analog Sensors)	Signal testing
Thermal Control System	Temperature Measurement Range (DS18B20)	-55°C to $+125^\circ\text{C}$	Datasheet
Thermal Control System	Measurement Accuracy	$\pm 0.5^\circ\text{C}$ (within -10°C to $+85^\circ\text{C}$)	Datasheet
Thermal Control System	Control Logic	$T < \text{Setpoint} \rightarrow \text{ON}$ / $T \geq \text{Setpoint} \rightarrow \text{OFF}$	Software testing
Thermal Control System	Control Type	Closed-loop control	System analysis
Thermal Control System	Water Heating Tank Capacity	≈ 0.5 Liter	Practical implementation

Thermal Control System	Heater Placement	Internal heater	Hardware inspection
Water Management System	Water Level Sensor	Analog (0–4095 ADC)	Reading verification
Water Management System	Pump Operation	Automatic ON at low level	Functional testing
Water Management System	Leak Detection	Conductivity-based detection	Water testing
Water Management System	Secondary Water Tank Capacity	≈0.5 Liter	Practical implementation
Water Management System	Water Supply Mechanism	Automatic refill from secondary tank	Functional testing
Control Unit	Microcontroller	ESP32 (Dual-core + Wi-Fi)	Datasheet + code verification
IoT System	Communication Type	Wi-Fi (Local Network)	Connectivity testing
IoT System	Data Update Rate	1–5 seconds	Application monitoring
IoT System	Application Features	Monitoring + Control + Alerts	App testing
Safety System	Dry-run Protection	Heater OFF when water level is LOW	Safety testing
Safety System	Thermal Control Safety	Heater OFF at Setpoint	Functional testing
Power System	Battery Type	12V Lead-Acid Rechargeable Battery	Hardware inspection
Power System	Battery Charging Source	Small Solar Panel	Hardware inspection
Power System	Operating Voltage	5V (control) / 12V (heater)	Electrical measurement

Power System	ESP32 Power Consumption	~80–260 mA	Datasheet
System Performance	Overall Efficiency	Up to 40% improvement using tracking	Literature comparison

1.5.3 Analytical Notes

The DS18B20 temperature sensor provides a measurement accuracy of approximately $\pm 0.5^{\circ}\text{C}$ within its operating range, which is sufficient for ON/OFF thermal control systems, resulting in a practical temperature stability of around $\pm 2^{\circ}\text{C}$.

The solar tracking accuracy of $\pm 5^{\circ}$ is appropriate for LDR-based systems, offering a balance between system simplicity and performance, while achieving an energy gain of approximately 20–40% compared to fixed systems.

The system operates using a 12V battery charged by solar energy, while the control circuitry operates at 5V, ensuring proper separation between power and control levels.

The system is designed for continuous operation (24/7 monitoring) with stable performance by minimizing oscillations in tracking movement and heating control through threshold-based decision logic.

The system design is based on integrating proven technologies from previous studies, including solar tracking systems [1], digital temperature sensing [2], automated water level control [3], and leakage detection mechanisms [4].

1.6 Project Contribution

This project presents a practical contribution through the development of an integrated system that combines solar tracking, thermal control, and Internet of

Things (IoT) technologies into a unified platform for residential applications. Unlike conventional systems that rely on separate functionalities, the proposed system integrates energy harvesting, heating control, and real-time monitoring within a single solution.

The project has evolved progressively through multiple stages. Initially, a single-axis solar tracking system was implemented during the semester project, followed by a dual-axis tracking system in Graduation Project 1 to improve tracking accuracy. In the current stage, the system has been extended to include a complete smart water heating application, incorporating user-defined temperature control and IoT-based remote monitoring.

The main contribution of this project lies in the effective integration of solar tracking and thermal heating within a single system, which enhances solar energy utilization and reduces reliance on conventional electrical energy. In addition, the system improves operational safety by incorporating protection mechanisms such as water leakage detection and dry-run protection, minimizing potential risks.

Furthermore, the system provides an interactive user interface that enables remote monitoring and control of water temperature through a mobile application, along with real-time alerts in case of abnormal conditions such as leakage or low water levels. This integration of monitoring and control contributes to reducing unnecessary heater operation, thereby minimizing energy waste and improving overall system efficiency.

Accordingly, this project represents an advanced step toward the development of smart solar water heating systems that combine efficiency, safety, and usability within a compact and cost-effective solution.

Unlike previous studies that focused on individual subsystems separately [1]–[4], this project presents an integrated system that combines solar tracking, thermal control, IoT monitoring, and safety mechanisms within a unified platform.

1.7 Design Methodology

A structured design methodology was adopted in this project, combining theoretical analysis with practical implementation to ensure the development of an efficient and integrated system that meets the defined performance requirements. Initially, a comprehensive theoretical study was conducted based on relevant research in solar tracking systems, thermal sensing, and control systems, in order to identify suitable approaches and select appropriate components.

Following the theoretical phase, the system was designed and implemented by dividing it into several subsystems, including the solar tracking system, thermal heating system, IoT monitoring system, and safety and protection system. This modular approach facilitated system development, allowing each subsystem to be analyzed and optimized independently before full system integration.

Component selection was based on criteria such as accuracy, efficiency, cost, and ease of integration. LDR sensors were used for solar tracking, servo motors for precise positioning, and a DS18B20 sensor for accurate and stable temperature measurement. The ESP32 microcontroller was selected due to its processing capabilities and built-in wireless communication, enabling the implementation of a smart and connected control system.

The control strategy is based on a comparison-driven decision logic. The solar panel orientation is determined by comparing LDR sensor readings, while the heating process is controlled based on a predefined temperature setpoint. The heater is activated when the temperature falls below the setpoint and deactivated once the

desired temperature is reached. In addition, safety mechanisms are triggered immediately in abnormal conditions such as water leakage or low water level.

The system was implemented using Arduino IDE for programming the ESP32, while Android Studio was used to develop the mobile application for remote monitoring and control. After implementation, the system was tested experimentally to evaluate its performance and stability. The results demonstrated correct system response under different operating conditions, achieving the intended functionality efficiently.

Table 1.2: System Components and Justification

Component	Reason for Selection
DS18B20	Provides high accuracy, measurement stability, easy integration, and suitability for smart control systems.
MG995 Servo Motor	Offers high torque, suitable speed, durability, ease of control, and wide availability.
LCD 16×2	Provides clear display, easy interfacing, real-time data updates, low cost, and ease of use.
LDR Sensor	Enables intelligent light tracking, simple installation, low cost, and sufficient accuracy for the application.
DC Voltage Meter	Ensures accurate and stable voltage monitoring, battery protection, and clear data visualization.
Water Level Sensor	Enhances system safety, prevents dry operation, and provides accurate water level monitoring.
Water Leakage Sensor	Enables instant leak detection, automatic protection, smart alerts, and fast response.
Water Valve	Ensures operational safety, protects system components, and provides immediate response to abnormal conditions.

BD243 Transistor	Allows safe control of the 12V heater, supports high current, ensures long-term stability, and integrates easily with microcontrollers.
Voltage Regulator	Provides stable voltage supply, protects electronic components, and ensures stable system performance.
Buzzer	Provides instant audible alerts, real-time response, and easy integration into the system.
ESP32 Microcontroller	Offers built-in Wi-Fi connectivity, high processing capability, multiple I/O pins, and flexible programming support.

The adopted design methodology is based on concepts validated in the literature, including solar tracking techniques [1], temperature sensing systems [2], water level control strategies [3], and safety mechanisms such as leakage detection [4].

1.8 Report Structure

This report is organized into five main chapters, in addition to appendices and references, ensuring a structured and systematic presentation of the project.

Chapter 1: General Introduction provides an overview of the project, its importance, objectives, basic concepts, and the design methodology adopted.

Chapter 2: Literature Review presents relevant studies and previous work related to the project, including a comparative analysis and identification of the research gap.

Chapter 3: System Analysis and Design describes the overall system architecture, components, and the mathematical modeling and engineering analysis used in the design process.

Chapter 4: Implementation and Testing focuses on the practical implementation of the system, along with experimental results, performance evaluation, and system validation.

Chapter 5: Conclusion and Recommendations summarizes the main achievements of the project, discusses the challenges encountered, and provides suggestions for future improvements and recommendations.

In addition, the report includes **appendices** containing detailed system designs, source code, and user manual, as well as a **references section** listing all scientific sources used in this work.

1.9 Chapter Conclusion

In this chapter, a comprehensive overview of the project has been presented, including its general concept, significance, problem statement, and objectives. The fundamental concepts and design methodology were also discussed, along with the project scope and technical requirements. Furthermore, the contribution of the project to the development of smart solar water heating systems has been highlighted.

These elements establish the theoretical and organizational foundation of the project and prepare for the next chapter, which focuses on reviewing existing studies and solutions in the field, in order to analyze them and utilize their findings in the development of the proposed system.

Chapter 2

Literature Review

2.1 Previous Studies in the Field

Recent advancements in solar energy and smart control systems have focused on improving energy efficiency, measurement accuracy, and system safety.

Study [1] presented a practical solar tracking system based on light sensors and mechanical actuation, demonstrating that solar tracking can increase power output by more than 30% compared to fixed systems by maintaining an optimal angle of solar radiation incidence.

In terms of thermal measurement, study [2] evaluated the DS18B20 sensor and showed strong agreement with reference sensors, with minimal deviation reaching $\pm 0.08^{\circ}\text{C}$ under stable conditions, confirming its suitability for accurate and stable temperature monitoring.

In the field of water management, study [3] proposed an Arduino-based automatic water level control system, demonstrating effective prevention of dry-run conditions and improved water management efficiency through automated sensor-based control.

Regarding safety systems, study [4] described a conductivity-based water leakage detection mechanism, where leakage is detected through the formation of an electrical path between electrodes, enabling fast and reliable response.

Overall, these studies focused on developing separate subsystems such as solar tracking [1], temperature sensing [2], water management [3], and safety systems [4], without integrating them into a unified smart system capable of real-time monitoring and control.

Table 2.1: Summary of Previous Studies

Study	Focus Area	Key Contribution	Limitation
[1] Solar Tracking System	Solar Energy	Increased power output >30%	No integration with heating or IoT
[2] DS18B20 Study	Temperature Measurement	High accuracy ($\pm 0.08^{\circ}\text{C}$)	Limited to sensing only
[3] Water Level Control	Water Management	Automatic pump control, dry-run protection	No energy system integration
[4] Leakage Detection	Safety System	Fast and reliable leak detection	Standalone function

2.2 Comparison with Existing Systems

Based on the reviewed studies, it is evident that most existing solutions focus on developing specific functionalities within isolated systems, without achieving full integration between different system components. Study [1] focused on improving energy efficiency through solar tracking, while study [2] addressed accurate and stable temperature measurement. In parallel, study [3] concentrated on water level management and dry-run prevention, whereas study [4] provided an effective solution for water leakage detection.

Although these solutions are effective within their respective domains, they lack integration between solar tracking, thermal control, water management, and safety

systems. Furthermore, they do not support real-time monitoring or remote control capabilities.

In contrast, the proposed system in this project offers a fully integrated solution that combines all these functionalities within a single platform. It incorporates dual-axis solar tracking to enhance energy efficiency [1], accurate temperature measurement using the DS18B20 sensor [2], automated water level control [3], and a leakage detection mechanism to ensure system safety [4].

Moreover, the system integrates Internet of Things (IoT) technologies, enabling real-time monitoring and remote control through a mobile application, along with local data visualization using an LCD display. This significantly enhances system usability and operational efficiency.

Therefore, the proposed system can be considered more comprehensive and advanced compared to existing solutions, as it does not focus on a single subsystem but rather provides an integrated approach combining efficiency, safety, and smart monitoring within one system.

Table 2.2: Comparison Between Existing Systems and Proposed System

Feature / System	[1] Solar Tracking	[2] Temperature Sensor	[3] Water Level Control	[4] Leak Detection	Proposed System
Solar Tracking	✓	✗	✗	✗	✓ (Dual-axis)
Temperature Measurement	✗	✓	✗	✗	✓
Water Level Control	✗	✗	✓	✗	✓

Leak Detection	×	×	×	✓	✓
IoT Monitoring	×	×	×	×	✓
Remote Control	×	×	×	×	✓
LCD Display	×	×	✓ (limited)	×	✓
System Integration	×	×	×	×	✓

2.3 Research Gap

Despite the significant advancements in solar energy systems and smart control technologies, previous studies have primarily focused on developing individual subsystems rather than integrated solutions. Solar tracking systems have demonstrated their ability to improve energy efficiency by maintaining an optimal angle of solar radiation incidence [1], while digital temperature sensors such as the DS18B20 have shown high accuracy and stability in temperature measurement [2].

In parallel, automated water level control systems have provided effective solutions for preventing dry-run conditions and improving water management [3], whereas leakage detection systems have enabled fast and reliable fault detection based on electrical conductivity principles [4].

However, despite their importance, these studies have addressed each subsystem independently, without achieving full integration between solar tracking, thermal control, water management, and safety systems within a single platform.

Furthermore, there is a clear lack of integration of Internet of Things (IoT) technologies in these systems, as most existing solutions do not support real-time monitoring or remote control, limiting their practical usability and user interaction.

Therefore, a clear research gap exists in the development of a fully integrated smart system that combines solar tracking, thermal control, water management, safety mechanisms, and IoT-based monitoring and control, aiming to improve system efficiency, reduce energy consumption, and ensure safe and reliable operation.

2.4 Theoretical Framework

The proposed project is based on fundamental theoretical concepts in control systems, solar energy, and Internet of Things (IoT) technologies. A simple and effective control methodology based on ON/OFF control is adopted, which is widely used in thermal systems due to its simplicity and reliability [2].

In the solar tracking subsystem, the theoretical approach is based on comparing light intensity using multiple LDR sensors placed around the solar panel. The system determines the optimal direction by comparing sensor readings and orienting the panel toward the highest light intensity, thereby improving energy capture efficiency, as demonstrated in light-based solar tracking systems [1].

From a thermal perspective, the system relies on basic heat transfer principles, where the heating process is controlled based on the difference between the current temperature and the desired setpoint. This is supported by a simplified energy model that relates thermal energy to heating time [5].

In terms of system components, the design is based on the integration of sensors, actuators, and microcontrollers. Sensors (such as LDR, DS18B20, water level, and leakage sensors) provide real-time data, while actuators (such as servo motors,

heater, and pump) execute control actions. The ESP32 microcontroller serves as the central processing unit that coordinates all system operations.

Additionally, the system incorporates IoT concepts by enabling wireless communication between the system and a mobile application via Wi-Fi, allowing real-time monitoring and remote control, which enhances system flexibility and usability.

Overall, the theoretical framework combines simplicity and effectiveness by employing appropriate control strategies and models that align with the system's design, ensuring a balance between performance, ease of implementation, and reliability.

2.5 Chapter Conclusion

This chapter reviewed a set of previous studies related to solar tracking systems, temperature sensing, water management, and safety mechanisms. The findings showed that solar tracking systems significantly improve energy utilization and increase power output [1], while digital sensors such as the DS18B20 provide high accuracy and stability in temperature measurement [2]. In addition, water level control systems effectively enhance water management and prevent dry-run conditions [3], whereas leakage detection systems enable fast and reliable responses to potential faults [4].

Despite these contributions, most studies focused on developing individual subsystems independently, without achieving full integration between system components. Furthermore, many of these solutions lack the integration of Internet of Things (IoT) technologies, limiting real-time monitoring and remote control capabilities.

Accordingly, there is a clear need for a fully integrated system that combines these functionalities within a unified platform, which is addressed in this project. Therefore, the next chapter focuses on the analysis and design of the proposed system based on the concepts and findings derived from the reviewed studies.

Chapter 3

System Analysis and Design

3.1 System Overview and Architecture

The proposed system represents an integrated solution for solar energy management and water heating, combining solar tracking, thermal control, water management, and safety mechanisms within a single platform based on an ESP32 microcontroller. This integration aims to improve solar energy utilization efficiency and ensure safe and reliable system operation, based on well-established principles such as solar tracking systems [1], accurate temperature sensing [2], water level control [3], and leakage detection mechanisms [4].

The system consists of multiple sensors that provide the required input data for decision-making. Four Light Dependent Resistors (LDRs) are used to detect solar radiation direction, while a DS18B20 sensor is employed for accurate water temperature measurement. In addition, a water level sensor monitors the tank condition, and a water leakage sensor detects abnormal conditions.

Based on these inputs, the ESP32 microcontroller executes a control algorithm to manage the system actuators. These include MG995 servo motors for dual-axis solar tracking, a heating element for temperature control based on the user-defined setpoint, a water pump and solenoid valve for water management, and a buzzer for safety alerts.

The system also provides two interfaces for monitoring and user interaction. A local LCD display presents real-time system data, including current temperature, battery voltage, and the user-defined temperature setpoint. Additionally, a mobile application enables remote monitoring and allows the user to set the desired temperature.

The system operates continuously in real time, where sensor data are periodically acquired, processed, and used to make control decisions, ensuring fast response to changing operating conditions.

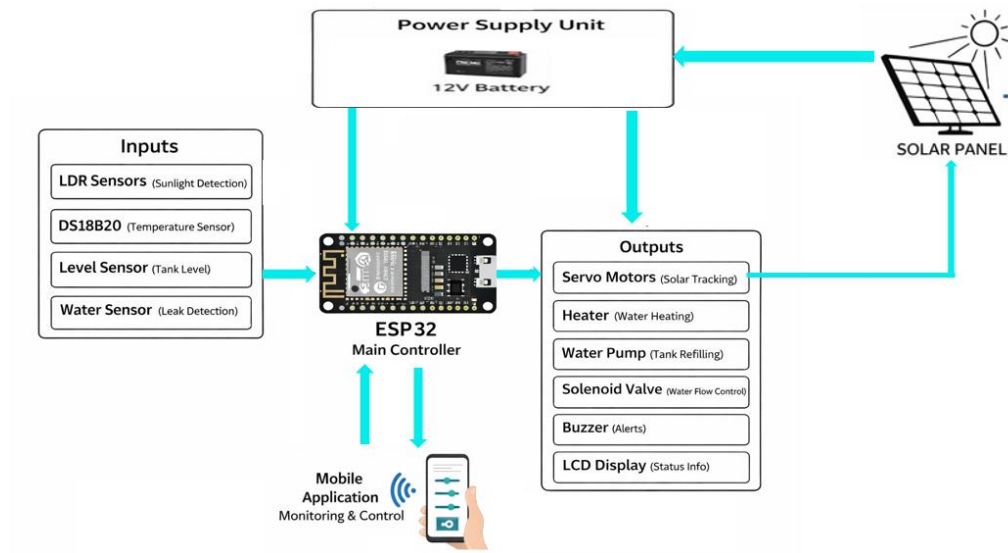


Figure 3.1: Block Diagram of the Proposed System

3.2 System Components

The proposed system consists of an integrated set of electronic and mechanical components that work together to achieve solar tracking, thermal control, water management, and safety functions. The selection of these components was based on criteria such as accuracy, reliability, efficiency, and ease of integration, following well-established concepts in previous studies, including solar tracking systems [1], accurate temperature sensing [2], water level control [3], and leakage detection mechanisms [4].

The system includes various sensors that provide essential input data for decision-making, such as LDR sensors for detecting solar radiation direction, a DS18B20

sensor for accurate and stable temperature measurement, as well as water level and leakage sensors to ensure safe operation. The ESP32 microcontroller serves as the central control unit, responsible for processing data and executing control algorithms.

At the actuator level, MG995 servo motors are used to enable dual-axis solar tracking, while a heating element controls the water temperature based on the user-defined setpoint. Additionally, a water pump and solenoid valve manage water flow, and a buzzer provides safety alerts during abnormal conditions. An LCD display is used for local data visualization, and a voltage monitoring system ensures stable power supply.

This integration of components results in a balanced system that combines high performance, safety, and user-friendly operation, making it suitable for smart residential applications.

Table 3.1: Detailed System Components and Selection Justification

Component	Category	Function	Reason for Selection
LDR (×4)	Sensor	Detect solar light intensity and direction	Low cost, simple design, sufficient accuracy for tracking [1]
DS18B20	Sensor	Measure water temperature	High accuracy, stability, digital output [2]
Water Level Sensor	Sensor	Monitor water level in tank	Prevent dry-run operation, ensure system safety [3]
Water Leakage Sensor	Sensor	Detect water leakage	Fast detection using conductivity principle [4]
DC Voltage Sensor	Sensor	Monitor battery voltage	Ensures stable power and system protection

ESP32	Control Unit	Process data and control system	Built-in Wi-Fi, high performance, flexibility
MG995 Servo Motor (×2)	Actuator	Control solar panel position	High torque, durability, suitable for tracking
Heater (12V)	Actuator	Heat water to setpoint	Efficient and easy control
Water Pump	Actuator	Transfer water between tanks	Reliable fluid control
Solenoid Valve	Actuator	Control water flow	Fast response and safe operation
Buzzer	Actuator	Provide alert signals	Immediate warning in fault conditions
LCD 16×2	Interface	Display system data	Real-time monitoring, easy interface

3.3 Control Strategy

The control strategy of the proposed system is based on a loop-based control algorithm implemented within the ESP32 microcontroller, where sensor data are continuously read, processed, and used to make real-time decisions. The strategy is designed to achieve four main functions: solar tracking, thermal control, water management, and safety protection, based on well-established principles from previous studies such as solar tracking systems [1], accurate temperature sensing [2], water level control [3], and leakage detection mechanisms [4].

The solar tracking system operates as a parallel process, where LDR sensor readings are continuously compared to determine the direction of maximum light intensity, and the servo motors are adjusted accordingly to orient the solar panel toward the sunlight. This approach improves solar energy capture efficiency, as demonstrated in solar tracking systems [1].

Within the main control flow, the system begins by reading all sensors, including battery voltage, temperature, water level, and leakage status. The battery voltage is used for monitoring purposes only and is displayed on both the LCD and the mobile application without directly affecting system operation decisions.

The system then checks for leakage conditions. If leakage is detected, immediate safety actions are triggered, including activating the buzzer, turning off the heater, controlling the pump and solenoid valve, and sending an alert to the mobile application. This mechanism is based on the conductivity principle used in leakage detection systems [4].

If no leakage is detected, the system proceeds to thermal control. The user-defined temperature setpoint is received from the mobile application and compared with the current water temperature measured using the DS18B20 sensor. If the measured temperature is below the setpoint, the heater is activated; otherwise, it is turned off, ensuring stable and accurate temperature regulation [2].

Additionally, the system verifies the water level before activating the heater. If the water level is low, the heater is turned off and alert mechanisms are activated to prevent dry-run conditions, following the principles of water level control systems [3].

At the end of each operation cycle, the LCD is updated and data are transmitted to the mobile application, including the current temperature, desired temperature, and battery voltage.

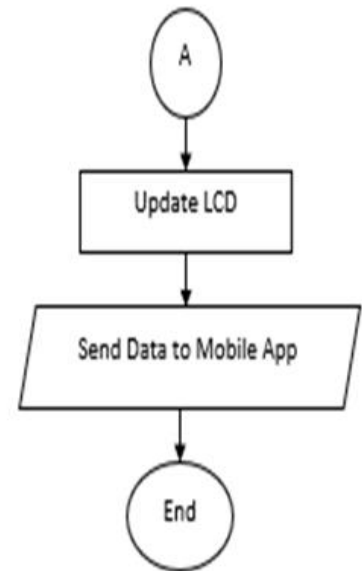
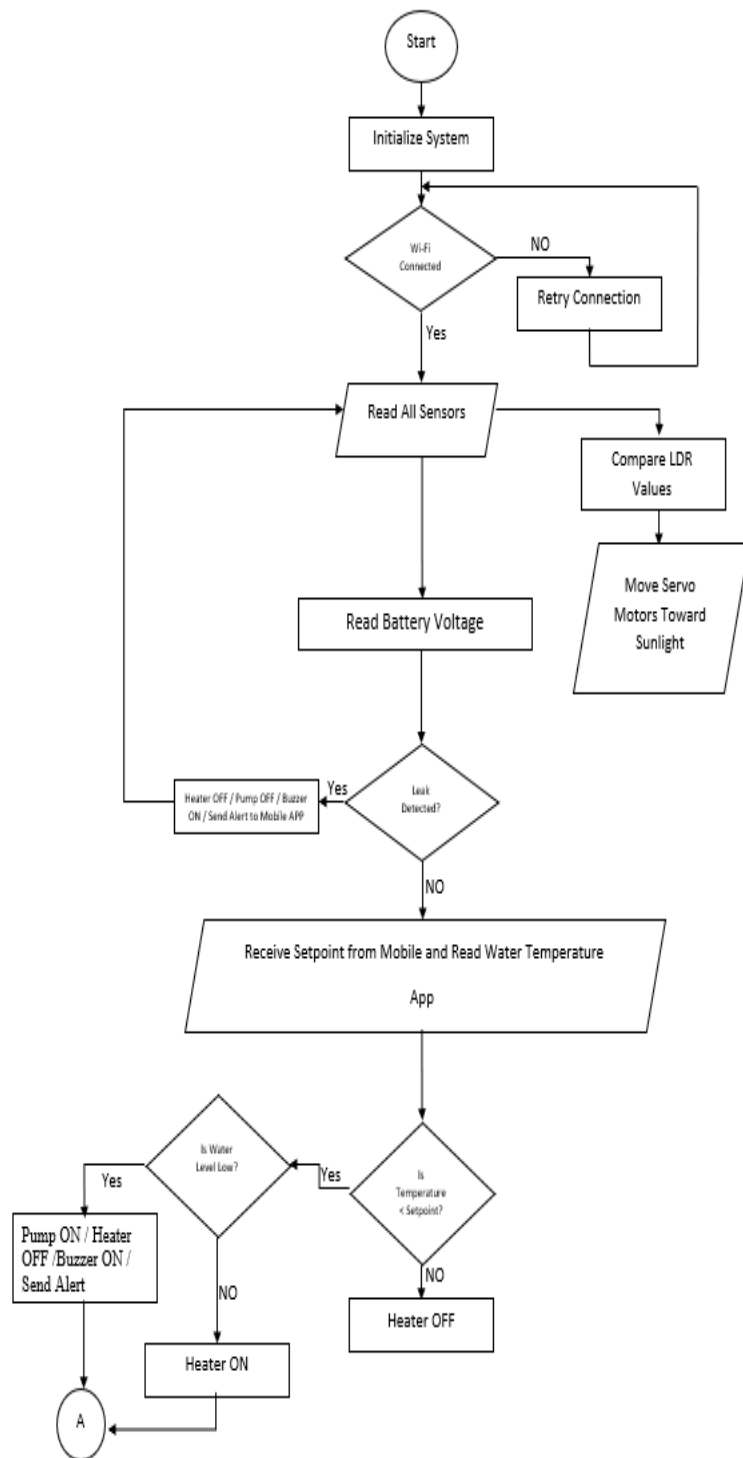


Figure 3.2: Flowchart of the Proposed System Control Strategy

3.4 Hardware Implementation

The proposed system is implemented using a hardware architecture centered around the ESP32 microcontroller, which acts as the main control unit. Sensors and actuators are interconnected according to the wiring diagram shown in Figure 3.3. This design ensures efficient integration of system components while maintaining stable power distribution and safe operation, based on established concepts such as solar tracking systems [1], temperature sensing [2], water management [3], and safety mechanisms [4].

The power system is based on a 12V battery charged by a solar panel. A DC-DC converter is used to step down the voltage to 5V for supplying the ESP32 and other low-voltage components. Sensors are powered directly from the ESP32 3.3V output, with all grounds connected in a common ground configuration to ensure signal stability and prevent measurement errors.

Four LDR sensors are connected to analog input pins (GPIO 32, 33, 34, 35) to detect light intensity from different directions. The DS18B20 temperature sensor is connected to GPIO 4 using the OneWire protocol, ensuring accurate temperature measurement [2]. The water level sensor is connected to GPIO 14, the leakage sensor to GPIO 39, and the voltage sensor to GPIO 36 for battery monitoring.

At the actuator level, two MG995 servo motors are connected to GPIO 25 and GPIO 26 to enable dual-axis solar tracking. High-power components such as the heater, water valve, and pump are controlled using driver circuits (e.g., BD243 transistors) connected to GPIO 19, GPIO 18, and GPIO 23, respectively, allowing safe switching of 12V loads. A buzzer is connected to GPIO 12 to provide audible alerts during fault conditions.

An I2C LCD is used for real-time data display, connected via SDA (GPIO 21) and SCL (GPIO 22), reducing wiring complexity. Low-voltage components such as servos, LCD, and pump are powered by 5V, while high-power loads such as the heater and valve operate at 12V.

Figure 3.3 illustrates the complete system wiring, including power distribution, sensor connections, and actuator interfacing, reflecting a well-integrated hardware design that ensures performance, safety, and ease of implementation.

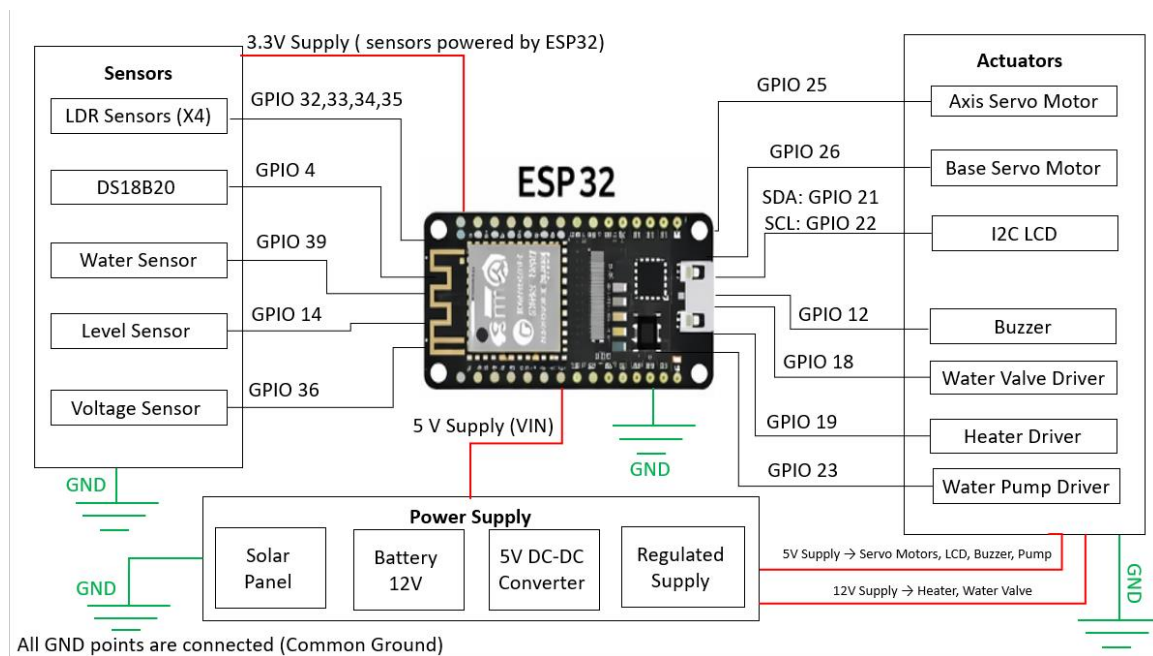


Figure 3.3: System Wiring Diagram

3.5 System Integration

The proposed system represents an integrated solution that combines hardware, software, and IoT communication within a unified operating environment based on the ESP32 microcontroller. The system software was developed using the Arduino IDE, utilizing several libraries such as Wi-Fi, OneWire, Dallas Temperature,

LiquidCrystal_I2C, and ESP32Servo to ensure seamless integration between system components.

The system operates using a continuous loop-based mechanism, where sensor data are read, processed, and used for decision-making within the loop() function. The update interval is approximately 200 milliseconds, ensuring fast response and near real-time data updates.

In terms of communication, the ESP32 operates as a server within a local Wi-Fi network, establishing a direct connection with the mobile application using socket-based communication over port 80. This setup allows the system to receive the user-defined temperature setpoint and continuously transmit system data such as current temperature and battery voltage, as implemented in the mobile application code.

The mobile application was developed using Android Studio and provides an interactive interface that allows the user to input the desired temperature, monitor real-time system parameters, and receive alerts in case of abnormal conditions such as water leakage or low water level. Data reception is handled using a dedicated background thread to maintain application responsiveness and ensure stable performance.

The system operates in a fully integrated manner, where all subsystems—including solar tracking, thermal control, water management, and safety mechanisms—run concurrently without conflict or delay. This integration enables the system to respond dynamically and efficiently to changing operating conditions.

This design reflects a high level of integration between hardware, software, and communication systems, representing a practical implementation of smart energy systems as discussed in previous studies [1][2][3][4].

3.6 System Testing and Operational Scenarios

Since the proposed system is based on a real implementation rather than a theoretical simulation model, an experimental testing approach was adopted to evaluate system performance under different operating conditions. A set of operational scenarios was conducted to validate solar tracking efficiency, thermal control accuracy, and safety system responsiveness, based on the principles established in previous studies [1][2][3][4].

In the thermal control scenario, the system was tested under low initial temperature conditions. A temperature setpoint was entered via the mobile application, and it was observed that the heater automatically activated when the measured temperature was below the setpoint. The heater remained active until the desired temperature was reached, after which it turned off, demonstrating stable and effective thermal control performance [2].

For the leakage detection scenario, a water leakage condition was simulated. The system responded immediately by activating the buzzer and sending an alert to the mobile application, confirming the fast response and reliability of the conductivity-based leakage detection mechanism [4].

In the low water level scenario, the water level inside the tank was reduced. The system successfully triggered an alert and automatically activated the pump to refill the tank, while preventing unsafe operation, demonstrating the effectiveness of the water management system in avoiding dry-run conditions [3].

Additionally, the solar tracking system was verified to operate continuously, adjusting the panel orientation in response to variations in light intensity, thereby improving solar energy capture efficiency as described in solar tracking systems [1].

All tests were conducted using real measured sensor data rather than simulated datasets, ensuring high reliability and accuracy of the results. The system demonstrated stable and integrated performance, with all subsystems operating concurrently without conflict, confirming its effectiveness in achieving efficiency, safety, and real-time responsiveness.

3.7 Mathematical Modeling and Design Calculations

The thermal design of the system is based on fundamental thermodynamic relations [5], where the water heating process is described by:

$$(3) Q = m * c * \Delta T$$

Where:

- Q is the required thermal energy
- m is the water mass
- c is the specific heat capacity
- ΔT is the temperature difference

The heating time is estimated using:

$$(4) t = Q \div P$$

Considering system losses, the actual heating time is given by:

$$(5) t_{\text{actual}} = Q \div \eta P$$

where η represents system efficiency.

These equations describe the relationship between water volume, temperature variation, and heating power, and are derived from standard heat transfer principles as presented in fundamental engineering references [5].

3.8 Power and Electrical Analysis

The proposed system is powered by a 12V battery that supplies electrical energy to all system components according to their operational requirements. The system includes low-power devices operating continuously, such as sensors and the control unit, in addition to higher-power components that operate only when required, such as the heater, pump, and servo motors.

This distribution helps reduce energy consumption and improve system efficiency, since not all components operate simultaneously at all times. For example, the heater only operates when water heating is required, while the pump operates only during low water level conditions. On the other hand, sensors and the control unit remain continuously active to monitor system status.

An approximate power analysis was conducted to estimate system power consumption, operating duration, and battery charging time using the solar panel.

The electrical power and energy calculations were based on fundamental electric circuit principles [6].

3.8.1 Electrical Power Calculation

This equation is used to calculate the electrical power consumed by each system component.

$$(6) P = V \times I$$

Where:

- P: Electrical Power (Watt)
- V: Voltage (Volt)
- I: Current (Ampere)

3.8.2 Battery Energy Calculation

This equation is used to estimate the total energy stored in the battery.

$$(7) E = V \times Ah$$

Where:

- E: Battery Energy (Wh)
- V: Battery Voltage (Volt)
- Ah: Battery Capacity (Ampere-hour)

3.8.3 Operating Time Calculation

This equation is used to estimate the approximate operating time of the system based on battery energy and consumed power.

$$(8) t = E / P$$

Where:

- t: Operating Time (hours)
- E: Battery Energy (Wh)
- P: Consumed Power (W)

3.8.4 Battery Charging Time Calculation

This equation is used to estimate the charging time of the battery using the solar panel.

$$(9) t = \text{Battery Capacity} / \text{Solar Panel Power}$$

Where:

- t : Charging Time
- Battery Capacity : Battery Energy

- Solar Panel Power : Solar Panel Power

Table 3.2: Estimated Electrical Power Consumption of System Components

Component	Voltage (V)	Current (A)	Power (W)
ESP32	5V	0.24A	1.2W
Servo Motors (×2)	5V	0.5A	5W
LCD Display	5V	0.03A	0.15W
DS18B20	3.3V	0.002A	0.006W
LDR Sensors	3.3V	0.001A	Negligible
Water Level Sensor	5V	0.02A	0.1W
Leakage Sensor	5V	0.02A	0.1W
Water Pump	5V	0.4A	2W
Heater	12V	1.5A	18W
Buzzer	5V	0.03A	0.15W

Assuming a battery capacity of approximately 12V / 7Ah, the total stored battery energy can be estimated as:

$$E = 12 \times 7 = 84 \text{ Wh}$$

Using a small solar panel with an approximate power of 15W, the theoretical charging time becomes:

$$t = 84 / 15 \approx 5.6 \text{ hours}$$

Considering practical losses and solar irradiance variations, the actual charging time can be estimated within approximately 7–9 hours of good sunlight exposure.

3.9 Communication Protocols

The system utilizes multiple communication protocols to ensure proper integration between its components. The I2C protocol is used to interface the LCD with the ESP32, reducing wiring complexity and simplifying hardware design.

The OneWire protocol is used for the DS18B20 temperature sensor, providing accurate temperature measurement using a single data line, making it suitable for precise thermal applications [2].

For external communication, the system relies on Wi-Fi technology, where the ESP32 operates as a server within a local network. Data exchange with the mobile application is achieved through socket-based communication, enabling real-time monitoring and remote control.

3.10 Chapter Conclusion

This chapter presented a comprehensive analysis and design of the proposed system, including the overall system architecture and key components. The control strategy and system operation were explained in detail, along with the hardware implementation and component interfacing.

System integration from both software and communication perspectives was also discussed, in addition to the testing methodology and operational scenarios. Furthermore, mathematical modeling and electrical analysis were introduced to provide a deeper understanding of system behavior and performance. Communication protocols were also presented to ensure proper interaction between system components.

Overall, this chapter demonstrated that the proposed system is based on an integrated design that balances efficiency, safety, and simplicity, providing a solid foundation for practical implementation and performance evaluation, which are discussed in the following chapter.

Chapter 4

Implementation and Testing

4.1 System Implementation

The proposed system was fully implemented in a real environment, where all electronic and mechanical components were assembled and interconnected according to the design specifications presented in the previous chapter. The system was tested as an integrated unit and demonstrated stable operation under real conditions.

Regarding the solar tracking system, the panel exhibited actual movement using servo motors, responding to variations in light intensity detected by LDR sensors. The system achieved an approximate tracking accuracy of around 80%, with minor fluctuations due to the analog nature of the measurements.

For the heating system, the heater is activated when a user-defined temperature setpoint is entered via the mobile application and operates until the desired temperature is reached, after which it turns off. If the temperature decreases afterward, the system does not automatically restart the heater; instead, it requires new user input. This reflects a user-triggered control approach.

In terms of communication, a stable connection was established between the ESP32 and the mobile application over a local Wi-Fi network, enabling real-time monitoring and control.

For safety and water management, both leakage and low water level scenarios were tested. In the case of leakage, the system activates the buzzer and sends an alert. When the water level drops, the system triggers an alert and automatically activates the pump to refill the tank, while preventing unsafe heater operation until normal conditions are restored. This confirms the effectiveness of both safety and water management mechanisms.

Finally, all subsystems—including solar tracking, water management, safety mechanisms, and the heating system under its defined control logic—operated simultaneously without conflict, demonstrating successful integration and stable system performance.

4.2 Experimental Results

The system performance was evaluated through a set of operational scenarios representing actual working conditions, aiming to verify tracking efficiency, safety system responsiveness, and communication stability between system components. Some results were obtained through direct practical testing, while certain thermal values were estimated based on theoretical calculations and the implemented tank capacity.

In the solar tracking subsystem, the system showed noticeable improvement in orienting the solar panel toward the light source compared to a fixed position. The servo motors continuously responded to changes in LDR sensor readings during operation, contributing to improved panel orientation throughout the experiment.

For the heating subsystem, heating performance was estimated using a water tank with an approximate capacity of 0.5 liters and a 12V heating element. Theoretical calculations indicated that increasing the water temperature from approximately 22°C to 40°C would require an estimated time ranging between 35 and 50 minutes under normal operating conditions. Figure (4.1) illustrates the approximate heating curve representing temperature variation over time during the heating process.

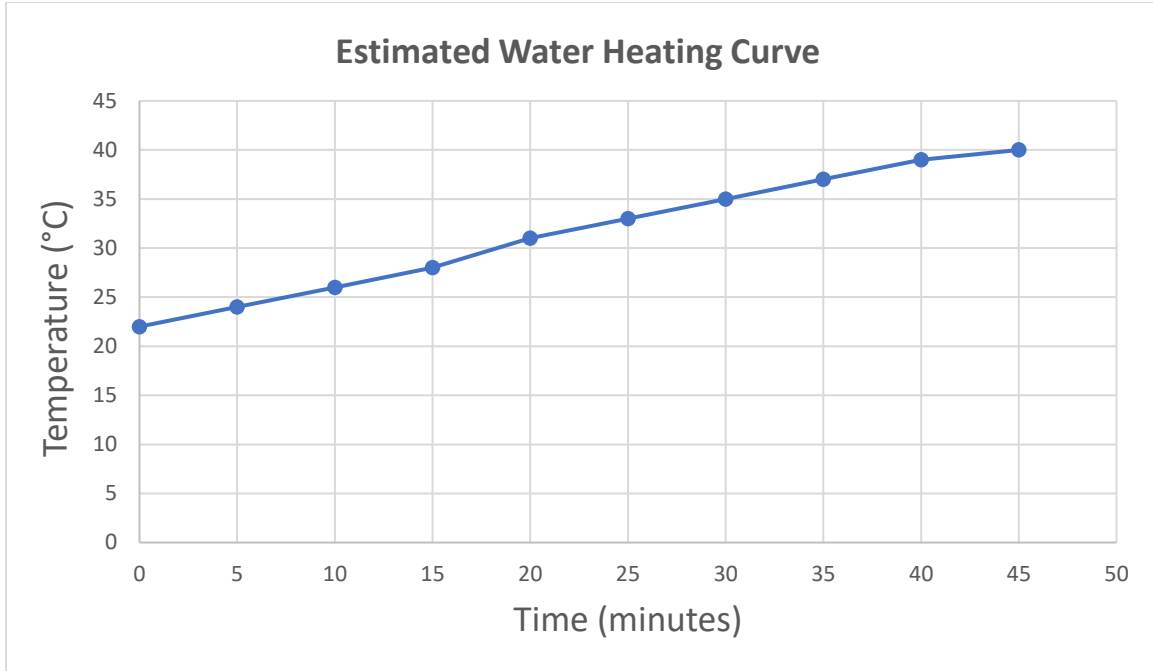


Figure 4.1: Estimated Water Heating Curve

Regarding battery performance, a gradual voltage drop was observed while operating different system loads, which is considered normal behavior for rechargeable battery-powered systems.

The safety mechanisms demonstrated fast response during practical testing, where alarms were activated within approximately one second in cases of water leakage or low water level. The pump was also automatically activated to refill the tank when necessary.

In terms of communication performance, the system achieved near real-time data transmission to the mobile application with a response time of approximately one second, enabling effective live monitoring during operation.

Overall, the experimental results demonstrated stable and integrated system operation, with good responsiveness under various operational conditions within the adopted design limits.

Table 4.1: Experimental Results of the Proposed System

Test Case	Initial Conditions	Action	Result	Response Time / Performance
Water Heating	$\approx 22^{\circ}\text{C} \rightarrow 40^{\circ}\text{C}$	Heater Activation	Gradual increase in water temperature	$\approx 35\text{--}50$ minutes (Estimated)
Solar Tracking	Variable light intensity	Servo Movement	Improved panel orientation toward light	Noticeable improvement during operation
Battery Voltage	$\approx 12\text{V}$ (Fully Charged)	System Operation	Gradual voltage drop under load	Normal decrease during operation
Water Leakage	Leakage detected	Protection Activation	Alarm + Notification	≈ 1 second
Low Water Level	Low water level	Pump Activation	Automatic tank refill	$\approx 1\text{--}2$ seconds
Communication and Updates	Application running	Data Transmission	Real-time value updates on mobile application	≈ 1 second

4.3 Performance Evaluation (KPIs)

The system performance was evaluated using a set of Key Performance Indicators (KPIs) to assess the achievement of design objectives, including tracking accuracy, system response time, thermal control performance, communication efficiency, and overall system reliability.

For solar tracking, the system achieved an approximate tracking accuracy of $\pm 5^{\circ}$, with a servo response time of approximately 1–2 seconds, which is acceptable for an LDR-based tracking mechanism. During operation, the system showed noticeable improvement in panel orientation toward the light source compared to a fixed position.

In the heating subsystem, the desired temperature was estimated to be achieved with a thermal stability of approximately $\pm 2^{\circ}\text{C}$ based on the implemented ON/OFF control logic and theoretical heating calculations.

In terms of monitoring and communication, the system demonstrated synchronized updates on both the LCD and mobile application with a response time of approximately 0.5 to 1 second, enabling effective real-time monitoring.

For safety mechanisms, both leakage detection and low water level response times were within approximately 1–2 seconds, with immediate alarm activation and automatic pump operation when required.

Regarding battery performance, the system demonstrated stable operation during testing, with gradual voltage reduction observed under electrical load conditions, which is considered normal for rechargeable battery-powered systems.

Finally, the system demonstrated reliable and stable operation, with all subsystems functioning concurrently without major failures or communication conflicts, confirming successful hardware and software integration.

Table 4.2: System Performance Indicators (KPIs)

Indicator	Approximate Value	Evaluation
Solar Tracking Accuracy	$\pm 5^{\circ}$	Good
Servo Response Time	1–2 seconds	Fast
Panel Orientation Performance	Noticeable improvement during operation	Effective
Thermal Control Accuracy	$\pm 2^{\circ}\text{C}$ (Estimated)	Stable

Data Update Rate	0.5–1 second	Near real-time
Leakage Detection Response	≈1 second	Fast
Low Water Level Response	1–2 seconds	Effective
Battery Performance	Gradual voltage drop under load	Normal
System Reliability	Continuous operation without major failure	High

4.4 Critical Analysis of the Project

This section presents a critical evaluation of the implemented system by analyzing its strengths and limitations in relation to the obtained performance indicators and practical testing results.

One of the main strengths of the system is the effective integration between all subsystems, including solar tracking, thermal control, water management, and application-based monitoring. These subsystems operated simultaneously without noticeable operational conflicts. This was reflected in the system performance indicators, such as the fast response of the safety mechanisms, where alarms and pump activation occurred within approximately 1–2 seconds. In addition, synchronized data updates were achieved on both the LCD and mobile application with a response time of approximately 0.5–1 second. The system also demonstrated noticeable improvement in solar panel orientation toward the light source during operation, contributing to enhanced overall system performance.

On the other hand, the solar tracking accuracy did not reach an ideal level, with an approximate accuracy of $\pm 5^\circ$, which is expected for systems based on LDR sensors due to their sensitivity to reflections, environmental lighting conditions, and

surrounding disturbances. In addition, the heating subsystem relies on an ON/OFF control mechanism, meaning that thermal regulation is not continuously maintained as in more advanced control systems. Furthermore, system operating duration depends on battery charge level and available solar irradiance during operation.

Compared to more advanced systems, the proposed system offers simplicity, low cost, and ease of implementation while integrating multiple functionalities into a single platform, including solar tracking, heating, IoT monitoring, and safety mechanisms. In contrast, some modern systems utilize more advanced control algorithms such as adaptive control or artificial intelligence techniques, which provide higher accuracy at the expense of increased complexity and cost.

Regarding future improvements, the system could be enhanced by implementing advanced control algorithms such as PID to improve tracking stability and thermal regulation. Cloud connectivity and smart home integration could also be added to achieve more flexible monitoring and control. In addition, predictive maintenance and energy consumption analysis techniques could be integrated in the future to improve overall system efficiency and reliability.

Overall, the proposed system achieves a good balance between performance, ease of implementation, and cost, while maintaining strong potential for future development toward more intelligent and efficient operation.

4.5 Cost Analysis and Final Evaluation

A preliminary cost analysis was conducted based on the approximate prices of the components available in the local market in order to evaluate the economic feasibility of the proposed system. Table (4.3) presents the estimated cost of the main components used in the project implementation, while considering market price variations.

Table 4.3: Estimated Cost of System Components (USD)

Component	Quantity	Estimated Cost (USD)
ESP32	1	12 – 18
Servo Motors (MG995)	2	18 – 30
LDR Sensors	4	1 – 3
DS18B20 Temperature Sensor	1	2 – 5
Water Level + Leakage Sensors	2	3 – 7
Water Pump	1	8 – 15
Heating Element	1	10 – 20
LCD Display (I2C)	1	5 – 10
12V Battery	1	25 – 50
Small Solar Panel	1	10 – 20
Miscellaneous Materials (Wiring, Frame, Connections)	—	15 – 35
Total Estimated Cost	—	109 – 213 USD

The overall system cost is considered reasonable for a prototype system integrating solar tracking, thermal control, safety mechanisms, and IoT monitoring within a single platform. In addition, the system utilizes relatively low-cost and widely available components, which simplifies implementation and future development.

Although the addition of solar tracking components and motors increases the initial system cost, these components contribute to improving solar energy utilization and enhancing the overall system performance during operation.

Experimental results demonstrated stable and effective system operation, successfully achieving the primary project objectives, including solar tracking, temperature control, water level monitoring, leakage detection, and mobile application monitoring. The system also shows strong potential for future development aimed at improving efficiency and reducing energy consumption using more advanced control techniques.

4.6 Chapter Conclusion

This chapter presented the practical implementation and performance evaluation of the proposed system. The system was successfully built and operated as an integrated unit under real conditions, demonstrating its ability to achieve the main objectives, including solar tracking, thermal control, and safety functions.

The experimental results were analyzed across different operating scenarios, confirming the system's effectiveness and reliability. Performance was further evaluated using Key Performance Indicators (KPIs), which demonstrated satisfactory response time, accuracy, and overall system stability.

A critical analysis was also conducted to highlight both the strengths and limitations of the system, reflecting a comprehensive understanding of its behavior and performance. In addition, a cost analysis was provided to assess the economic feasibility of the system.

Overall, this chapter confirms the successful implementation and operation of the proposed system, providing a solid basis for the final conclusions and future recommendations presented in the next chapter.

Chapter 5

Conclusion and Recommendations

5.1 Project Summary and Key Achievements

This project presents an integrated smart system that combines dual-axis solar tracking and water heating, along with safety and monitoring functions, within a single platform based on the ESP32 microcontroller. The system aims to improve solar energy utilization efficiency while ensuring safe and reliable operation.

All main project objectives were successfully achieved. The solar tracking system effectively improves panel orientation and energy capture efficiency, while the heating system is capable of reaching the desired temperature. In addition, a Wi-Fi-based communication system was implemented to enable remote monitoring and control, along with safety mechanisms including water leakage detection and water level monitoring.

The proposed system is distinguished by integrating multiple functions (tracking, heating, monitoring, and safety) into a relatively simple and cost-effective design compared to more advanced systems, while maintaining acceptable performance, in line with concepts presented in previous studies .

The project demonstrates that it is possible to develop a practical and integrated system that balances efficiency, responsiveness, and ease of implementation, confirming the success of the design and its potential for real-world applications and future improvements.

5.2 Challenges and Difficulties

The implementation of the project involved several technical and practical challenges, which are typical in multi-component integrated systems. From a technical perspective, one of the main challenges was related to the accuracy of the solar tracking system. The use of LDR sensors, which rely on analog measurements,

makes the system sensitive to environmental factors such as light variation and reflections, resulting in minor fluctuations in tracking accuracy, as reported in solar tracking studies

In the thermal control system, an ON/OFF control approach was adopted due to its simplicity and reliability. However, this method may not maintain perfectly stable temperature under all conditions, which is expected for this type of control system

On the practical side, challenges were encountered in component wiring and system organization, especially due to the integration of multiple sensors and actuators. Proper wiring and grounding were essential to ensure signal stability and avoid measurement errors. Additionally, managing different voltage levels (3.3V, 5V, and 12V) required careful design to ensure safe and stable operation.

Furthermore, the availability of components in the local market and price variations posed organizational challenges, along with limited time for extensive experimental testing, which led to focusing primarily on core system functionalities.

Overall, these challenges contributed to a deeper understanding of the system and helped refine its design, leading to a balanced solution that combines performance, simplicity, and practical feasibility.

5.3 Future Work

The proposed system can be significantly enhanced through several technical improvements aimed at increasing performance, accuracy, and overall intelligence. One of the main areas of improvement is the solar tracking system, where higher-precision sensors or improved signal processing algorithms can be used to reduce environmental noise effects and enhance tracking accuracy.

The thermal control system can also be improved by replacing the basic ON/OFF control with more advanced control algorithms such as Proportional-Integral-Derivative (PID) control, which can provide better temperature stability and reduce fluctuations around the desired setpoint.

From a communication perspective, the system can be integrated with cloud-based platforms to enable extended remote monitoring and control, as well as data logging and long-term analysis. Artificial intelligence techniques can also be applied to optimize energy consumption and support intelligent decision-making based on real operating data.

Another important enhancement is the implementation of predictive maintenance, where sensor data can be analyzed to detect potential faults before they occur, thereby improving system reliability. Additionally, smart scheduling can be introduced based on user behavior and environmental conditions such as solar radiation and ambient temperature.

Finally, the system can be integrated with smart home platforms, allowing it to operate within a broader intelligent environment and enabling centralized control.

Overall, these improvements provide a clear pathway for transforming the system from a functional prototype into a more advanced, intelligent, and scalable solution.

5.4 Recommendations

Based on the experience gained from this project, it is recommended that students working on similar systems focus on proper planning and system design, particularly in terms of component organization and wiring, as this plays a critical role in ensuring system stability and reducing operational errors. It is also advisable

to test each subsystem independently before full integration, allowing for easier troubleshooting and performance optimization.

From a practical perspective, selecting reliable and suitable components is essential, especially for analog-based systems such as LDR sensors, as component quality directly affects system accuracy. Proper energy management and battery selection are also crucial to ensure continuous and stable operation, particularly in solar-powered systems.

From a technical standpoint, implementing more advanced control algorithms in future stages can significantly improve system stability and efficiency. Additionally, proper documentation of all development stages is highly recommended to support future improvements.

Overall, this project demonstrates that success is not only determined by the idea itself, but by effective implementation, practical understanding, and the ability to handle real-world challenges.

5.5 Chapter Conclusion

This chapter concluded the overall project by summarizing the key achievements, discussing the challenges encountered during implementation, and presenting future development opportunities and practical recommendations. It reflects a comprehensive understanding of all project stages, from design to implementation and evaluation.

In general, the project demonstrates the feasibility of developing an integrated smart system that combines solar tracking, thermal control, and safety mechanisms within a single platform, while maintaining a balance between performance, cost, and

simplicity. It also highlights the importance of such systems in supporting renewable energy utilization and smart system applications in real-life scenarios.

This work provides a solid foundation for future research and development, whether through improving system accuracy, enhancing control strategies, or expanding application scope. Therefore, the project concludes with a practical and scientific framework that opens the door for further innovation and advancement in this field.

Appendices

The appendices include supporting materials for the report, including source codes, and the user manual, in order to fully document the design and implementation stages.

Appendix A: Source Code

This appendix includes the source codes developed for the system implementation, including the ESP32 firmware and the mobile application code.

A.1 ESP32 Main Control Code

The following code represents the main ESP32 firmware used to control solar tracking, heating, safety mechanisms, LCD display, and Wi-Fi communication.

Listing A.1: ESP32 Main Control Code

```
#include <WiFi.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <OneWire.h>

#include <DallasTemperature.h>

#include <ESP32Servo.h>

const char* ssid = "YAZAN";

const char* password = "*****";

IPAddress local_IP(192, 168, 1, 33);
```

```

IPAddress gateway(192, 168, 1, 1);

IPAddress subnet(255, 255, 255, 0);

WiFiServer server(80);

WiFiClient client;

#define ONE_WIRE_BUS 4 \\ PIN 4 FOR DS18B20

String data_rec;

OneWire oneWire(ONE_WIRE_BUS);

DallasTemperature sensors(&oneWire);

LiquidCrystal_I2C lcd(0x27,16,2);

Servo myservo_1, myservo_2;

int pos_1 = 85, pos_2 = 180;

int data_1, data_2, data_3, data_4;

int movee = 0, wanted = 0;

float volt, temp;

void setup() {

Serial.begin(9600);

if (!WiFi.config(local_IP, gateway, subnet)) {

Serial.println("STA Failed to configure");

```

```

}

WiFi.begin(ssid, password);

Serial.print("Connecting to WiFi");

while (WiFi.status() != WL_CONNECTED) {

  delay(500);

  Serial.print(".");

}

Serial.println("\nConnected!");

Serial.println(WiFi.localIP());

server.begin();

pinMode(14, INPUT_PULLUP); \\ PIN 14 FOR LEVEL SENSOR

pinMode(12, OUTPUT); \\ PIN 12 FOR BUZZER

pinMode(18, OUTPUT); \\ PIN 18 FOR Water valve Driver

pinMode(19, OUTPUT); \\ PIN 19 FOR Heater Driver

pinMode(23, OUTPUT); \\ PIN 23 FOR Water PUP Driver

analogSetAttenuation(ADC_11db);

sensors.begin();

Wire.begin(21,22); \\ PUV 21 FOR SDA , PIN 22 FOR SCL

```

```

lcd.begin();

lcd.backlight();

myservo_1.attach(25); \\ PIN 25 FOR Axis Servo Motor

myservo_1.write(pos_1);

myservo_2.attach(26); \\ PIN 26 FOR Base Sereo Motor

myservo_2.write(pos_2);

}

void loop() {

WiFiClient newClient = server.available();

if (newClient) {

client = newClient;

Serial.println("Client connected");

}

if (client && client.connected() && client.available()) {

data_rec = client.readStringUntil('\n');

wanted = data_rec.toInt();

lcd.clear();

}

```

```
if ((digitalRead(14) == HIGH) && (digitalRead(12) == LOW)) { \\ PIN 14 FOR  
LEVEL SENSOR AND PIN 12 FOR BUZZER
```

```
if (client && client.connected()) client.println("c");
```

```
digitalWrite(12, HIGH); \\ PIN 12 FOR BUZZER
```

```
digitalWrite(18, HIGH); \\ PIN 18 FOR Water valve Driver
```

```
digitalWrite(23, HIGH); \\ PIN 23 FOR Water PUP Driver
```

```
}
```

```
if ((digitalRead(14) == LOW) && (digitalRead(23) == HIGH)) { \\ PIN 14 FOR  
LEVEL SENSOR AND PIN 23 FOR Water PUP Driver
```

```
digitalWrite(23, LOW); \\ PIN 23 FOR Water PUP Driver
```

```
}
```

```
int watterr = analogRead(39); \\ PIN 39 FOR Water Sensor
```

```
if ((watterr > 500) && (digitalRead(12) == LOW)) { \\PIN 12 FOR BUZZER
```

```
if (client && client.connected()) client.println("a");
```

```
digitalWrite(18, HIGH); \\ PIN 18 FOR Water valve Driver
```

```
digitalWrite(12, HIGH); \\ PIN 12 FOR BUZZER
```

```
}
```

```
if ((watterr < 500) && (digitalRead(12) == HIGH) && (digitalRead(14) ==  
LOW)) { \\ PIN 12 FOR BUZZER AND PIN 14 FOR LEVEL SENSOR
```



```

if (client && client.connected()) client.println("d");

digitalWrite(12, LOW); \\ PIN 12 FOR BUZZER

digitalWrite(18, LOW); \\ PIN 18 FOR Water valve Driver

}

int adcValue = analogRead(36); \\ PIN 36 FOR Voltage Sensor

volt = (float)adcValue * 3.3 / 4095.0;

volt = volt * 3.9;

sensors.requestTemperatures();

temp = sensors.getTempCByIndex(0);

if (client && client.connected()) {

client.println("V");

client.println(volt);

client.println("T");

client.println(temp);

}

delay(200);

if (temp >= wanted) {

digitalWrite(19, LOW); \\ pIN 19 FOR Heater Driver

```

```

} else {

digitalWrite(19, HIGH); \\ pIN 19 FOR Heater Driver

}

lcd.setCursor(0,0);

lcd.print("now");

lcd.setCursor(0,1);

lcd.print(temp);

lcd.setCursor(6,0);

lcd.print("want");

lcd.setCursor(6,1);

lcd.print(wanted);

lcd.setCursor(11,0);

lcd.print("volt");

lcd.setCursor(11,1);

lcd.print(volt);

data_1 = analogRead(33); \\ PIN 33 FOR LDR1

data_2 = analogRead(35); \\ PIN 33 FOR LDR2

if (data_1 > 2800) myservo_1.write(70);

```

```

if (data_2 > 2800) myservo_1.write(145);

data_3 = analogRead(32); \\ PIN 33 FOR LDR3

data_4 = analogRead(34); \\ PIN 33 FOR LDR4

if ((data_3 > 2500) || (data_4 > 2500)) {

if (movee == 0) {

movee = 1;

pos_2 = (pos_2 == 180) ? 80 : 180;

myservo_2.write(pos_2);

}

}

else {

movee = 0;

}

```

A.2 Mobile Application User Interface Code

The following code represents the user interface layout of the mobile application developed using Android Studio. It allows the user to set the desired temperature, monitor current temperature and voltage, and control the system via Wi-Fi connection.

Listing A.2: Mobile Application UI Code

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#0B1C2C"
    tools:context=".MainActivity">
```

```
<TextView

    android:id="@+id/title"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="SMART CONTROL PANEL"
    android:textColor="#00E5FF"
    android:textStyle="bold"
    android:textSize="22sp"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="24dp"/>
```

```
<ToggleButton

    android:id="@+id/conn"
    android:layout_width="70dp"
```

```
android:layout_height="70dp"
android:background="@drawable/wifi_off"
android:textOn=""
android:textOff=""
android:onClick="onToggleClicked_1"
app:layout_constraintTop_toBottomOf="@id/title"
app:layout_constraintEnd_toEndOf="parent"
android:layout_marginTop="16dp"
android:layout_marginEnd="16dp"/>
```

```
<androidx.cardview.widget.CardView
    android:id="@+id/card1"
    android:layout_width="0dp"
    android:layout_height="100dp"
    app:cardCornerRadius="16dp"
    app:cardBackgroundColor="#142B3D"
    app:layout_constraintTop_toBottomOf="@id/conn"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_margin="12dp">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal"
    android:gravity="center_vertical"
```

```
android:padding="16dp">
```

```
<TextView
```

```
    android:layout_width="0dp"  
    android:layout_height="wrap_content"  
    android:layout_weight="1"  
    android:text="TEMP SET"  
    android:textColor="#FFFFFF"  
    android:textSize="18sp" />
```

```
<EditText
```

```
    android:id="@+id/EditTextttt"  
    android:layout_width="80dp"  
    android:layout_height="50dp"  
    android:background="#1E3A52"  
    android:gravity="center"  
    android:text="00"  
    android:textColor="#00E5FF"  
    android:textSize="22sp"  
    android:inputType="number"  
    android:maxLength="3"  
    android:cursorVisible="true" />
```

```
</LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
<androidx.cardview.widget.CardView
    android:id="@+id/card2"
    android:layout_width="0dp"
    android:layout_height="100dp"
    app:cardCornerRadius="16dp"
    app:cardBackgroundColor="#142B3D"
    app:layout_constraintTop_toBottomOf="@id/card1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_margin="12dp">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:padding="16dp">
```

```
<TextView
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="CURRENT TEMP"
    android:textColor="#FFFFFF"
    android:textSize="18sp" />
```

```

<TextView
    android:id="@+id/noww"
    android:layout_width="80dp"
    android:layout_height="50dp"
    android:background="#1E3A52"
    android:gravity="center"
    android:text="30"
    android:textColor="#00E5FF"
    android:textSize="22sp" />

</LinearLayout>
</androidx.cardview.widget.CardView>

<androidx.cardview.widget.CardView
    android:id="@+id/card3"
    android:layout_width="0dp"
    android:layout_height="100dp"
    app:cardCornerRadius="16dp"
    app:cardBackgroundColor="#142B3D"
    app:layout_constraintTop_toBottomOf="@id/card2"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_margin="12dp">

<LinearLayout
    android:layout_width="match_parent"

```



```
android:layout_height="match_parent"
android:orientation="horizontal"
android:gravity="center_vertical"
android:padding="16dp">
```

```
<TextView
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="VOLTAGE"
    android:textColor="#FFFFFF"
    android:textSize="18sp" />
```

```
<TextView
    android:id="@+id/voltt"
    android:layout_width="80dp"
    android:layout_height="50dp"
    android:background="#1E3A52"
    android:gravity="center"
    android:text="12"
    android:textColor="#00E5FF"
    android:textSize="22sp" />
```

```
</LinearLayout>
</androidx.cardview.widget.CardView>
```

<Button

```
    android:id="@+id/sendd"
    android:layout_width="0dp"
    android:layout_height="60dp"
    android:text="SEND DATA"
    android:textStyle="bold"
    android:textSize="18sp"
    android:textColor="#FFFFFF"
    android:backgroundTint="#00E5FF"
    app:layout_constraintTop_toBottomOf="@id/card3"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_margin="16dp"/>
```

<TextView

```
    android:id="@+id/showww"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:text="SYSTEM READY"
    android:gravity="center"
    android:textColor="#00E5FF"
    android:textSize="18sp"
    android:textStyle="bold"
    app:layout_constraintTop_toBottomOf="@id/sendd"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
```

```
        android:layout_margin="12dp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

A.3 Mobile Application Logic & Communication Code

The following code represents the main logic of the mobile application, including Wi-Fi communication with the ESP32, data transmission, and real-time data reception.

Listing A.3: Mobile Application Logic and Communication Code

```
package com.example.myapplication;

import android.os.Bundle;

import android.os.Handler;

import android.os.Looper;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;
```

```
import java.io.BufferedReader;

import java.io.IOException;

import java.io.InputStream;

import java.io.InputStreamReader;

import java.io.OutputStream;

import java.net.InetSocketAddress;

import java.net.Socket;


public class MainActivity extends AppCompatActivity {


    String SERVER_IP = "192.168.1.33";

    int SERVER_PORT = 80;


    Socket socket;

    OutputStream outputStream;

    InputStream inputStream;


    Thread workerThread;
```

```
volatile boolean stopWorker;
```

```
EditText EditTextttt;
```

```
TextView nowww, volt, showww;
```

```
ToggleButton conn;
```

```
int tt = 0, vv = 0;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    Button senddd = findViewById(R.id.sendd);
```

```
    conn = findViewById(R.id.conn);
```

```
    showww = findViewById(R.id.showww);
```

```
    nowww = findViewById(R.id.noww);
```

```
    volt = findViewById(R.id.volt);
```

```
EditTextttt = findViewById(R.id.EditTextttt);
```

```
senddd.setOnClickListener(v -> {
```

```
    if (socket == null || !socket.isConnected() || outputStream == null) {
```

```
        showww.setText("Not Connected ✕");
```

```
        return;
```

```
    }
```

```
    String msg = EditTextttt.getText().toString().trim();
```

```
    if (msg.isEmpty()) {
```

```
        showww.setText("Enter a value first ⚠");
```

```
        return;
```

```
    }
```

```
    new Thread(() -> {
```

```
        try {
```

```

byte[] data = (msg + "\n").getBytes("UTF-8");

outputStream.write(data);

outputStream.flush();

runOnUiThread() -> showww.setText("Sent: " + msg + " ");

} catch (IOException e) {

    e.printStackTrace();

    runOnUiThread() -> showww.setText("Send Error: " +
e.getMessage() + " ");

}

}).start();

});

}

public void onToggleClicked_1(View v) {

    if (connn.isChecked()) {

```

```

new Thread(() -> {

    try {

        socket = new Socket();

        socket.connect(new InetSocketAddress(SERVER_IP,
SERVER_PORT), 3000);

        outputStream = socket.getOutputStream();

        inputStream = socket.getInputStream();

        runOnUiThread(() -> {

            connn.setBackgroundResource(R.drawable.wifion);

            showww.setText("Connected ");

        });

        beginListenForData();

    } catch (Exception e) {

        e.printStackTrace();

        runOnUiThread(() -> {

```



```

        connn.setChecked(false);

        connn.setBackgroundResource(R.drawable.wifioff);

        showww.setText("Connection Failed: " + e.getMessage());

    });

}

}).start();

} else {

    new Thread(() -> {

        try {

            closeConnection();

        } catch (IOException e) {

            e.printStackTrace();

        }

        runOnUiThread(() -> {

            showww.setText("Disconnected");

            connn.setBackgroundResource(R.drawable.wifioff);

        });

```

```

    }).start();

}

}

```

```

void beginListenForData() {

    Handler handler = new Handler(Looper.getMainLooper());

    stopWorker = false;

    workerThread = new Thread(() -> {

        try {

            BufferedReader reader = new BufferedReader(

                new InputStreamReader(inputStream, "UTF-8"));

            while (!stopWorker && socket != null && socket.isConnected()) {

                String data = reader.readLine();

                if (data == null) break;

                String finalData = data.trim();

```

```

handler.post() -> {

    if (finalData.equals("V")) {

        vv = 1;

    } else if (finalData.equals("T")) {

        tt = 1;

    } else if (vv == 1) {

        vv = 0;

        volttt.setText(finalData);

    } else if (tt == 1) {

        tt = 0;

        nowww.setText(finalData);

    } else if (finalData.equals("a")) {

        showww.setText("Water leak 🚰");

    } else if (finalData.equals("c")) {

        showww.setText("Low water level ⚠️");

    } else if (finalData.equals("d")) {

        showww.setText("-----");
    }
}

```

```

        }

    });

}

} catch (IOException e) {

    runOnUiThread() -> {

        showww.setText("Disconnected  ");

        connn.setChecked(false);

        connn.setBackgroundResource(R.drawable.wifi0ff);

    });

}

});

workerThread.start();

}

void closeConnection() throws IOException {

    stopWorker = true;

```

```

        if (workerThread != null) {

            workerThread.interrupt();

            workerThread = null;

        }

        try { if (outputStream != null) outputStream.close(); } catch (IOException
ignored) {}

        try { if (inputStream != null) inputStream.close(); } catch (IOException
ignored) {}

        try { if (socket != null) socket.close(); } catch (IOException ignored)
{}

        outputStream = null;

        inputStream = null;

        socket = null;

    }

    @Override

    protected void onDestroy() {

```

```

        super.onDestroy();

        new Thread(() -> {

            try { closeConnection(); } catch (IOException ignored) {}

        }).start();

    }

}

```

Appendix B: User Manual

This appendix provides the basic user manual for operating the system, including startup steps, mobile application connection, temperature setting, and safety monitoring.

B.1 System Startup

1. Ensure that the 12V battery is properly connected.
2. Verify that all sensors and actuators are correctly wired.
3. Power on the system and wait for the ESP32 to connect to the Wi-Fi network.
4. Confirm that system data appears on the LCD display.

B.2 Mobile Application Connection

1. Open the mobile application.
2. Press the Wi-Fi connection button.
3. Once connected, the connection status will be displayed.
4. The application will begin showing real-time temperature and battery voltage.

B.3 Temperature Setting

1. Enter the desired temperature in the **TEMP SET** field.
2. Press the **SEND DATA** button.
3. The value will be sent to the ESP32.
4. The heater will operate until the target temperature is reached, then it will stop.

B.4 Safety Monitoring

- In case of water leakage, an alarm will be activated and a warning message will appear in the application.
- In case of low water level, an alarm will be triggered and the pump will automatically refill the tank.
- The system continuously displays temperature and voltage on both the LCD and the mobile application.

B.5 Shutdown Procedure

1. Disconnect from the mobile application.
2. Turn off the system by disconnecting the 12V power supply.
3. Ensure that the heater and pump are turned off before shutting down the system.

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